

N'GUIGMI, Niger

WMO: 610490

Lat: 14.250N

Lon: 13.117E

Elev: 938

StdP: 14.20

Time Zone: 1.00 (E01)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 53.0	(c) 55.3	(d) 17.7	(e) 13.9	(f) 83.7	(g) 19.4	(h) 15.1	(i) 80.7	(j) 19.9	(k) 71.1	(l) 17.9	(m) 74.4	(n) 0.4	(o) 40	(p) 0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	27.4	110.5	70.3	108.9	70.3	107.0	70.4	79.5	94.0	78.6	93.1	78.0	92.4	4.3	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.6	143.6	83.4	75.6	138.7	83.1	74.7	134.5	83.2	43.8	92.7	42.9	93.2	42.1	92.3	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 15.7	(b) 12.7	(c) 10.0	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(4) 15.7	(5) 12.7	(6) 10.0	(7) 38.2	(8) 83.8	(9) 1.0	(10) 2.6	(11) 37.5	(12) 85.7	(13) 36.9	(14) 87.2	(15) 36.3	(16) 88.7	(17) 35.6	(18) 90.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	84.4	70.7	76.3	83.3	91.4	93.8	93.4	89.3	85.9	88.5	87.7	79.8	72.2		
	DBStd	8.69	5.18	6.06	5.10	3.81	2.97	3.30	3.91	3.45	2.95	2.87	4.32	4.86		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	11	8	1	0	0	0	0	0	0	0	0	0	2		
	CDD50	12544	640	737	1031	1242	1359	1302	1217	1111	1154	1168	894	689		
	CDD65	7080	183	318	566	792	894	852	752	646	704	703	444	226		
	CDH74	99006	2107	3846	7737	12270	14198	13197	10545	8120	9525	9753	5262	2446		
	CDH80	61191	895	2030	4717	8428	9929	8975	6322	4221	5653	6087	2893	1041		
(14) Wind	WSAvg	2.8	3.3	3.3	3.6	2.9	2.3	2.6	2.9	2.1	1.8	2.6	2.9	3.0		
(15) Precipitation	PrecAvg	7.6	0.0	0.0	0.0	0.0	0.2	0.6	2.2	4.0	1.1	0.1	0.0	0.0		
	PrecMax	19.4	0.0	0.0	0.2	0.8	0.8	3.0	6.8	12.1	4.2	1.8	0.0	0.0		
	PrecMin	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.2	0.0	0.0	0.0	0.0		
	PrecStd	3.9	0.0	0.0	0.0	0.1	0.2	0.6	1.5	2.2	1.1	0.3	0.0	0.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.4	104.1	108.0	112.4	113.0	110.9	107.2	102.9	106.2	105.9	101.9	97.1		
		MCWB	62.6	65.0	67.1	69.9	70.5	71.6	74.2	76.9	71.3	70.7	68.0	64.1		
	2%	DB	92.9	100.1	104.7	110.1	110.9	108.8	104.1	99.1	103.9	104.3	99.1	92.9		
		MCWB	60.3	63.4	65.4	69.5	70.2	72.7	75.4	77.3	71.9	69.8	65.8	60.8		
	5%	DB	89.1	96.0	101.8	108.2	109.2	106.9	101.5	97.0	102.0	102.7	96.4	89.6		
		MCWB	58.6	61.9	64.5	68.6	70.0	73.3	75.7	77.1	73.5	69.3	64.3	59.4		
	10%	DB	84.9	91.6	98.7	105.9	107.1	104.7	98.9	94.7	99.7	100.5	93.6	86.3		
		MCWB	57.1	60.2	62.9	67.8	70.2	73.8	75.7	76.8	74.3	69.0	63.0	58.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.3	68.0	72.0	76.7	78.8	80.1	79.9	80.6	80.4	79.3	75.8	65.3		
		MCDB	94.5	97.6	101.0	104.0	99.2	98.8	94.8	92.3	95.4	95.7	95.1	96.2		
	2%	WB	61.3	65.1	68.4	73.9	77.0	78.3	78.4	79.3	78.8	77.0	69.8	62.2		
		MCDB	90.3	95.6	97.8	101.3	100.2	98.3	93.9	91.2	93.3	93.1	93.1	90.2		
	5%	WB	59.4	62.8	66.0	72.0	75.8	77.3	77.7	78.7	77.9	75.5	66.0	60.5		
		MCDB	87.5	93.7	97.4	100.5	98.8	97.5	92.9	90.5	91.9	91.1	92.5	87.3		
	10%	WB	57.4	60.8	64.0	70.1	74.7	76.5	77.1	77.9	77.1	73.9	63.8	58.8		
		MCDB	83.8	90.6	96.5	99.4	97.3	96.4	92.1	89.5	90.7	90.5	90.8	84.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	28.8	29.4	29.3	29.3	27.4	23.5	19.2	17.5	22.7	26.9	27.8	27.6		
		MCDBR	33.4	33.8	33.1	31.5	30.7	26.3	22.0	20.3	26.0	28.8	29.8	30.6		
		MCWBR	14.7	13.9	13.1	11.9	11.3	9.0	6.5	4.8	7.7	10.1	12.2	12.8		
	5% WB	MCDBR	33.0	32.3	31.4	27.9	24.1	22.1	19.2	18.1	21.7	25.7	28.9	29.7		
		MCWBR	15.3	14.7	13.7	12.6	9.4	7.8	5.9	5.3	6.9	10.3	14.0	13.4		
(40) Clear-Sky Solar Irradiance	taub		0.596	0.650	0.696	0.741	0.732	0.727	0.707	0.659	0.659	0.681	0.597	0.587		
	taud		1.603	1.497	1.431	1.384	1.414	1.472	1.563	1.647	1.622	1.517	1.589	1.614		
	Ebn at Noon		223	219	214	205	203	201	207	219	219	209	221	220		
	Edh at Noon		82	94	103	108	103	96	88	82	84	91	82	79		
(44) All-Sky Solar Radiation	RadAvg		1807	1967	2120	2201	2200	2108	1974	1918	2032	1994	1867	1747		
	RadStd		49	44	91	77	88	85	61	83	47	56	44	43		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page